Final Thesis Presentation
On

"Exploring the Reasons Behind Land Cover Change and Evaluating the Management Strategies of a Selected Protected Area of Bangladesh"

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Level 4/ Term 1

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Date: 23 June 2024

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Outline of the Presentation

1. Research Background
2. Title & Objectives
3. Rationale
4. Methodology
5. Causes of Land Cover Change
6. Management Effectiveness
7. Major Findings & Recommendation

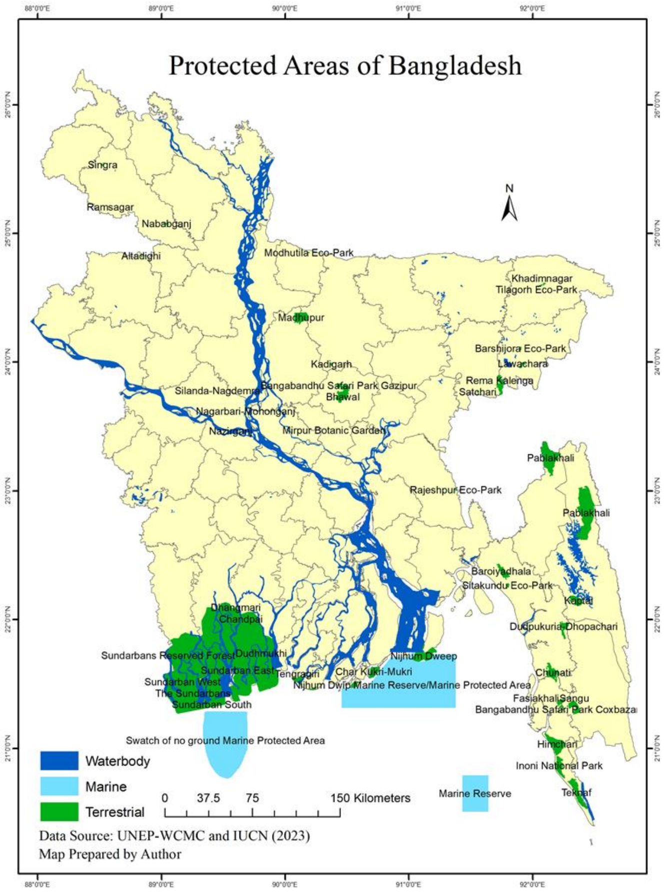
At present, **30 percent** of total tropical, subtropical and oceanic **forest areas** are legally **declared protected areas** (FAO, 2020). **Protected forest areas** continue to degrade and **lose forest cover** (Weisse et al., 2023).

Forest cover constitutes for **only 14.1%** of the total land area in Bangladesh (FAO, 2020).

To conserve the forest areas, the **Bangladesh Government** first **initiated** the process of **declaring protected areas** back **in 1960** (BFD, 2024).

Until now, **53 protected areas** have been declared in Bangladesh, covering **469871.613 hectares land** areas, representing **3.18% area** of the country (BFD, 2024).

Bangladesh had **2.22 million hectares** of natural forest in year **2010**, covering 16% of total land area; where, **17.8 kilo hectares** of natural forest land were **lost in year 2023** (GFW, 2023).



Thesis Title & Objectives

Thesis Title: Exploring The Reasons Behind Land Cover Change and Evaluating the Management Strategies of A Selected Protected Area of Bangladesh

Objectives

1. To explore the reasons behind land cover changes of a selected protected area in Bangladesh
2. To evaluate management strategies for selected protected area in Bangladesh.

Rationale

- **Protected areas** of Bangladesh are constantly **converted into agricultural lands, settlements**; making the country's conservation effort more challenging (Masum et al., 2023; Rahman et al., 2023).
- **Previous studies** in Bangladesh only concentrated on **identifying the extent of land cover change** in protected areas through **quantitative** method (Appendix).
- Understanding **specific context** and identifying the **under lying causes of land cover change** are crucial to effectively manage the protected areas (Twongyirwe et al., 2018).
- Till date many management strategies were undertaken by Bangladesh Forest Department for conserving the areas, however the **continuing deforestation question** the existing **management approach**.
- For formulating **adaptive strategies** to **address the threats** and vulnerabilities of the protected areas, first it is important to **assess the effectiveness** of present **management strategies** (Hockings, 2006).

Importance of Protected Area Management in Global Perspective

- ❑ **Aichi Target 11**
  (Strategic Plan for Biodiversity 2011-2020)

Protected areas should be effectively managed, ecologically representative, well-connected, and integrated into broader landscapes and seascapes.

Source: Convention on Biological Diversity, 2010

- ❑ **Sustainable Development Goals**

Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt biodiversity loss

Source: United Nations, 2015

Methodology

Problem Identification

Formulation of Objective

Selection of Study Area

Data Collection

Data analysis

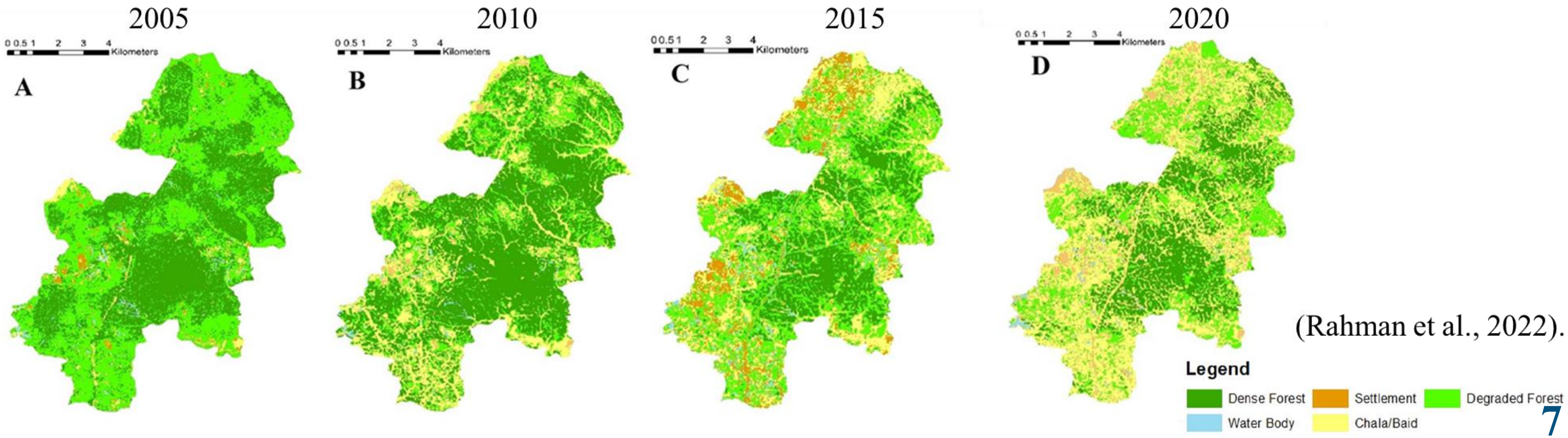
Formulation of Recommendation

- In **1982**, Bhawal forest was gazetted as a national park under the administrative control of Bangladesh Forest Department. **36% Sal forest** cover existed **in 1985**, only **10% forest cover is remained**, 0.12 million hectres of land (BFD, 2023).

- **Bhawal Sal forest** is one of the **most threatened** protected area because of increased human pressure on land (Ahsan et al., 2016).

- **Bhawal National Park** possess some **distinctive characteristics**.
  - **Connected** with an important national highway (**N3-Dhaka-Mymensingh Highway**)
  - Only **at 40 kilometers distant** from the capital **Dhaka** city.

- Rahman et al. (2023) showed that the **dense forest cover of Bhawal Sal** forest has **decreased by 50%** from 2005 to 2020.



(Rahman et al., 2022).

Methodology

Problem Identification

Formulation of Objective

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Data analysis

Formulation of Recommendation

Primary Data Collection		
Method	Involved persons	Discussed Topics
Key informant Interview (KII)	▪ Forest officials, forest managers ▪ Officers from 3 organizations (**CNRS, IUCN, Arranyak Foundation**) who work in the protected areas. ▪ **Total 19** KIIs	▪ Context of the study area, its history ▪ Reasons for forest depletion ▪ Key problems and threats ▪ Challenges in forest managements ▪ Scope for improvement
Focus Group Discussion (FGD)	5 FGDs with local people and indigenous people	▪ Temporal changes in the study area ▪ Livelihood practice of the participants ▪ Key problems of the study area

❑ **Secondary Data Collection** from Bangladesh Forest Department:
Amount of encroached land, number of industries, and other relevant data

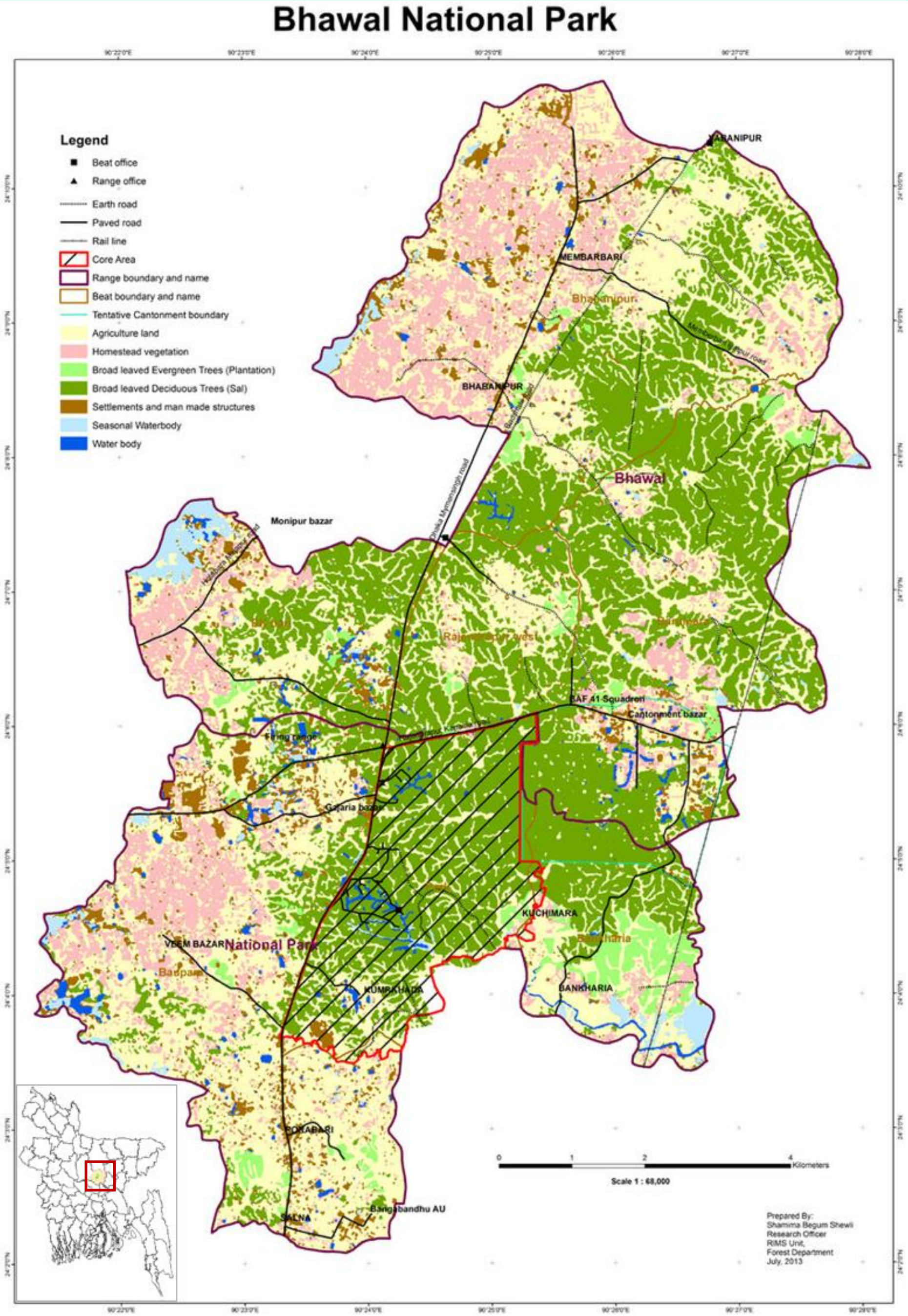
Methodology

- Problem Identification
- Formulation of Objective
- Selection of Study Area
- Data Collection
- Data analysis
- Formulation of Recommendation

PRA tools	Purpose	Date
Social and Resource Mapping	Identify existing social and natural resources	11 January 2024
Transect Walk	A transect walk was conducted along the Dhaka-Mymensingh highway within the study area to depict cross sectional view	11 January 2024
Historical Timeline	Identify the chronological events and temporal changes in study area	18 January 2024
Trend Analysis	To identify the change in livelihood and impact on forest resources	18 January 2024
Pair Wise Rankings	The priority problems were ranked behind forest land cover change	11 February 2024
Venn Diagram	The venn diagram is prepared to identify the relevant stakeholders with the forest management, both within and outside the forest area.	29 February 2024

- **Data Triangulation:** A preliminary analysis is conducted to **identify the common information** and **check the inconsistencies** in the data. The author **cross-checked the data and categorized** them.

Study Area Profile



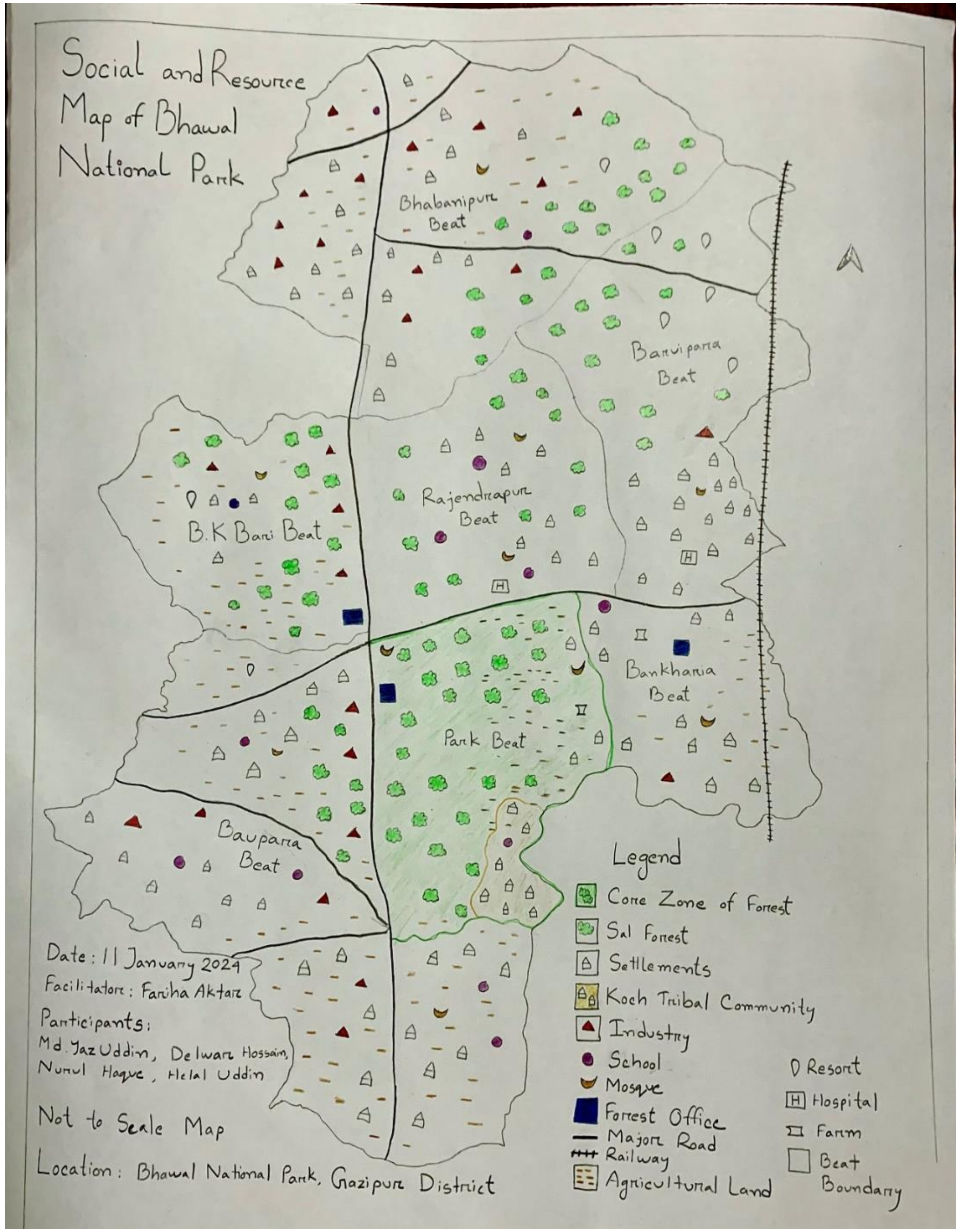
Source: BFD, 2016

Bhawal National Park		
Forest Range	Forest Beats	Area (in Acre)
1. National Park Forest range	1. Park Beat	2498.181
	2. Bonkhoria	825.314
	3. Boupara	1448.2531
2. Bhawal Forest Range	4. Bhabanipur	2600.9746
	5. B.K. Bari	1029.1715
	6. Rajandrapur	2254.7875
	7. Baroipara	2246.8803

Source: BFD, 2016

Study Area Profile

❑ Social and Resource Map of BNP

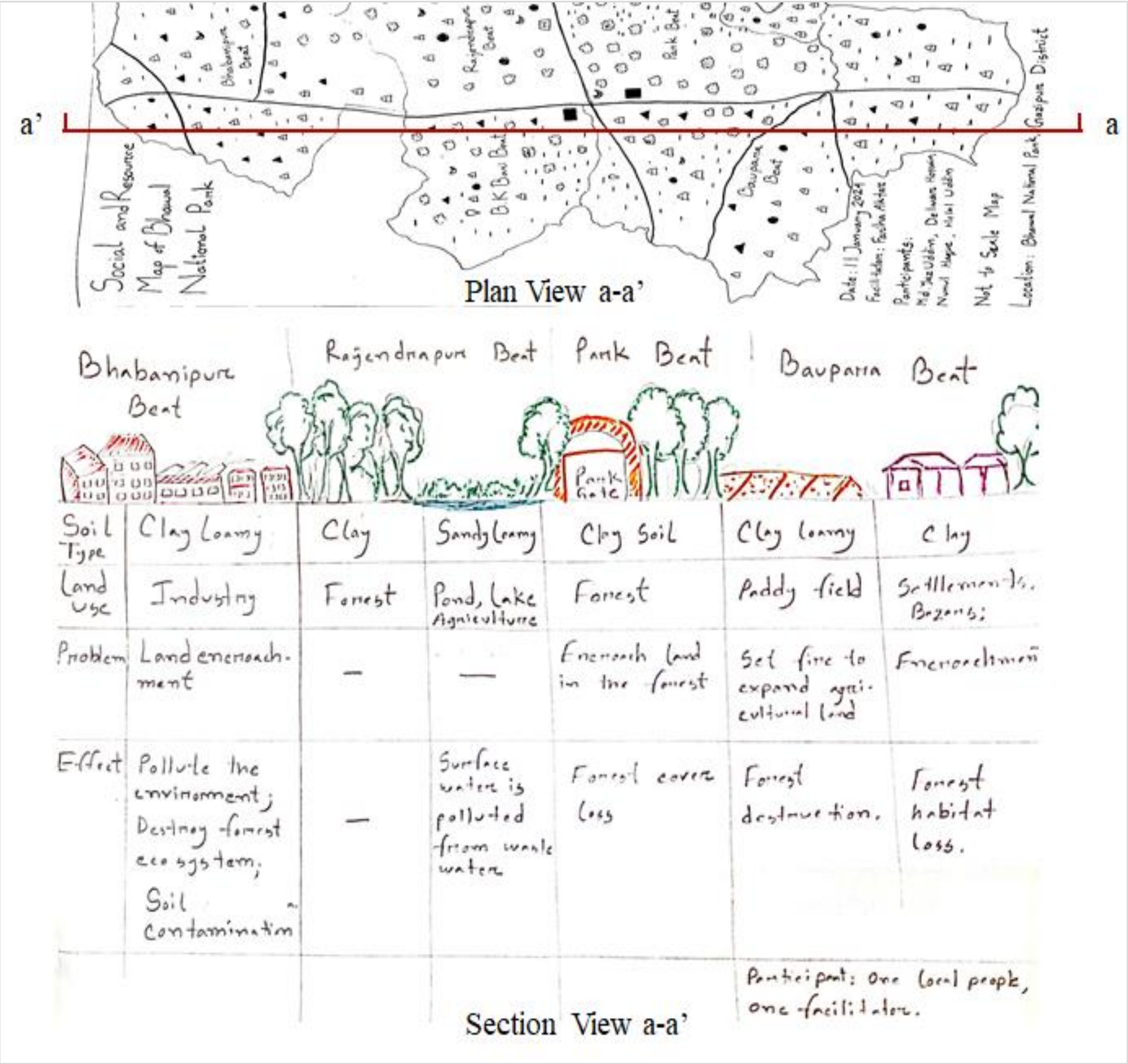


Resource Type	Number
Forest Offices	Forest Beat Office: 03, ACF Office: 01
Visitor Facilities	Rest House: 06, Cottage: 13
Watch Tower	02
Staff Facility	ACF Quarter: 01, Beat Officer Quarter: 01
	Forest Guard Barrack: 01
Patrolling Vehicles	Motorcycle: 02, Pickup: 01, Truck: 01
Educational Institution	Government Primary School: 39 Non Govt. High and Junior High School: 21. College: 05
Religious Institution	Mosque: 198, Temple: 22, Church: 01
Hospital	02
Community Clinic	21
Private Resorts	25

Source: Field Survey, 2024

Study Area Profile

- Cross Sectional View of the Study Area



Source:
Field Survey, 2024

Study Area Profile

❑ Temporal Change in the Study Area

Timeline	Incidence
Before 1950	**Bhawal Zamindar** ruled the area. **Koch Tribal** has been living in BNP for **100 years**
1980	**Dhaka-Mymensingh** Highway (N3) Construction
1982	**Declaration** of Bhawal forest as **a protected area**. Sal forest restoration programme was undertaken.
1988	**After** the massive **flood**, **industries** started to **shift** in this area for having **high elevation**
1994	**Establishment** of primary **school** in **Koch community**
1995	There were **60-70 households** in the indigenous community.
2000-2010	**Development** activities started, **construction of road networks**, **electricity lines, water lines**. The **forest** land **cover** was greatly **destroyed** during this period.
2024	Around **200 households** in **Koch** community.

Source: Prepared by Author (2024) from KII and FGD

Source: Field Survey, 2024

Source: Field Survey, 2024

Objective 01

To explore the reasons behind land cover changes of a selected protected area in Bangladesh

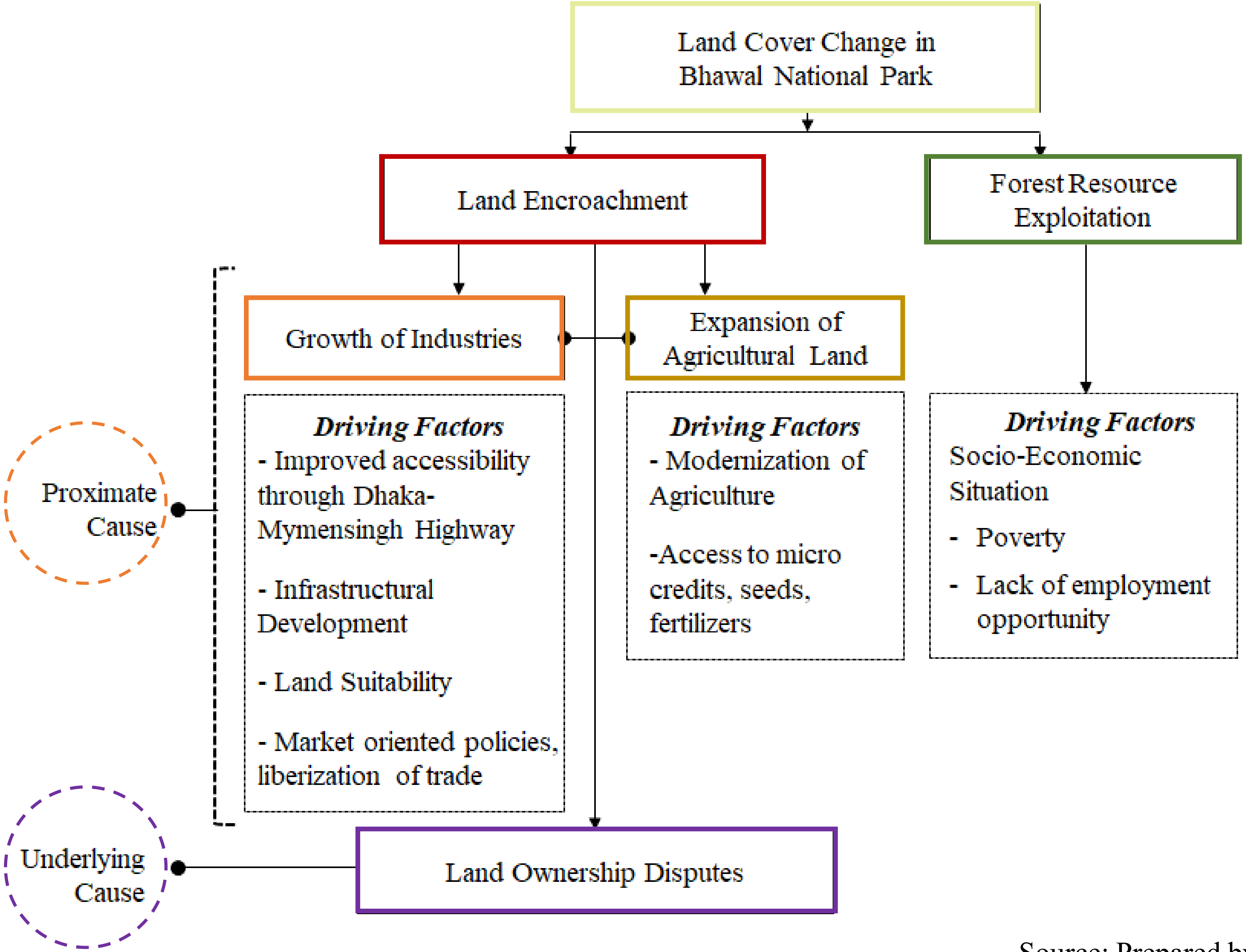
Objective 01: To explore the reasons behind land cover changes of a selected protected area in Bangladesh

❑ Pair Wise Matrix for Identifying the Major Causes Behind Land Cover Change

Causes of Land Cover Change in BNP	LE	ID	AG	LD	FR	IND	Frequency	**Rank**
Land encroachment (LE)		LE	LE	LD	LE	LE	3	II
Infrastructural development (ID)			AG	LD	ID	IND	1	V
Expansion of agricultural land (AG)				LD	AG	IND	2	IV
Land ownership disputes (LD)					LD	LD	5	I
Forest resource exploitation (FR)						IND	0	VI
Industrialization (IND)							4	III

Source: Prepared by Author (2024) from KII and FGDs

Objective 01: To explore the reasons behind land cover changes of a selected protected area in Bangladesh



Source: Prepared by Author (2024) from KII and FGDs

Objective 01: To explore the reasons behind land cover changes of a selected protected area in Bangladesh

❑ Description of the Reasons for Forest Land Cover Change in the Study Area

Land Ownership Disputes Regarding the Forestlands

Timeline	Incidence
1940	**Total lands** of Bhawal was **recorded** through **Cadastral Survey**
Before 1950	Under the **feudal system**, Bhawal landlords possessed the lands.
1950	The East Bengal State Acquisition and Tenancy Act **broke the feudal system**. SA Khatiyan was prepared.
1950-1956	**Court of Wards** retained the lands
1956	Total **recorded land** in **CS** survey was **entitled to BFD**
1961-1965	**Revisional survey (RS)** was conducted, **faulty recorded** lands **under individual** names.
1999	Record correction cases was filed by BFD

Source: Prepared by Author (2024) from KII

❑ Description of the Reasons for Forest Land Cover Change in the Study Area

Land Encroachment Scenario in Bhawal National Park

Land in BNP	Area (in Acre)
Total Encroached Land	**989.36** (Land Value = 197872 crore BDT)
Recovered Land	**326.28** (Only **30%** of encroached land)
Encroached by Industries	**151.44**
Encroached by Cottage, Agriculture	40.43

Source: BFD, 2024

Encroached Land by Different Land Uses in National Park Range

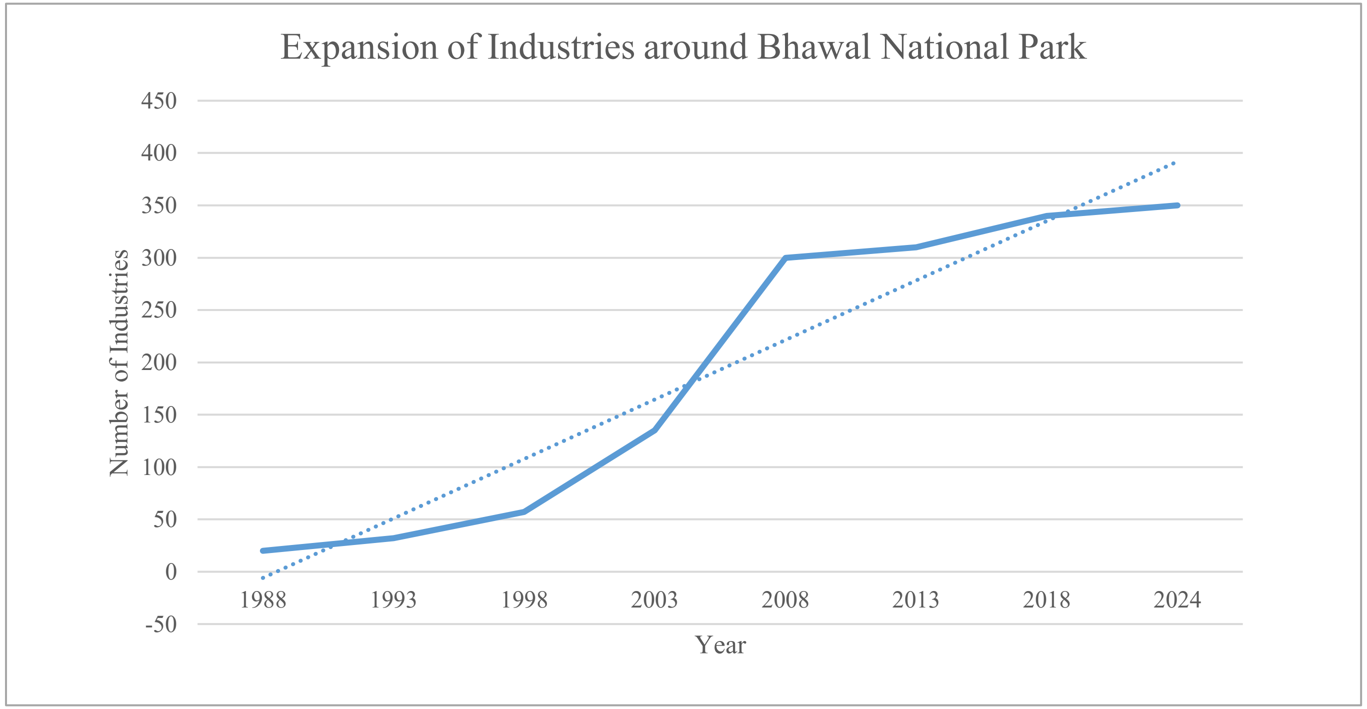
Land Use	Area (in Acre)
Permanent Infrastructure, Industries	23.72
Bazar, Resorts, Cottage	17.26
Settlements	137.95
Agriculture	16.15
Total	195.08

Source: BFD, 2024

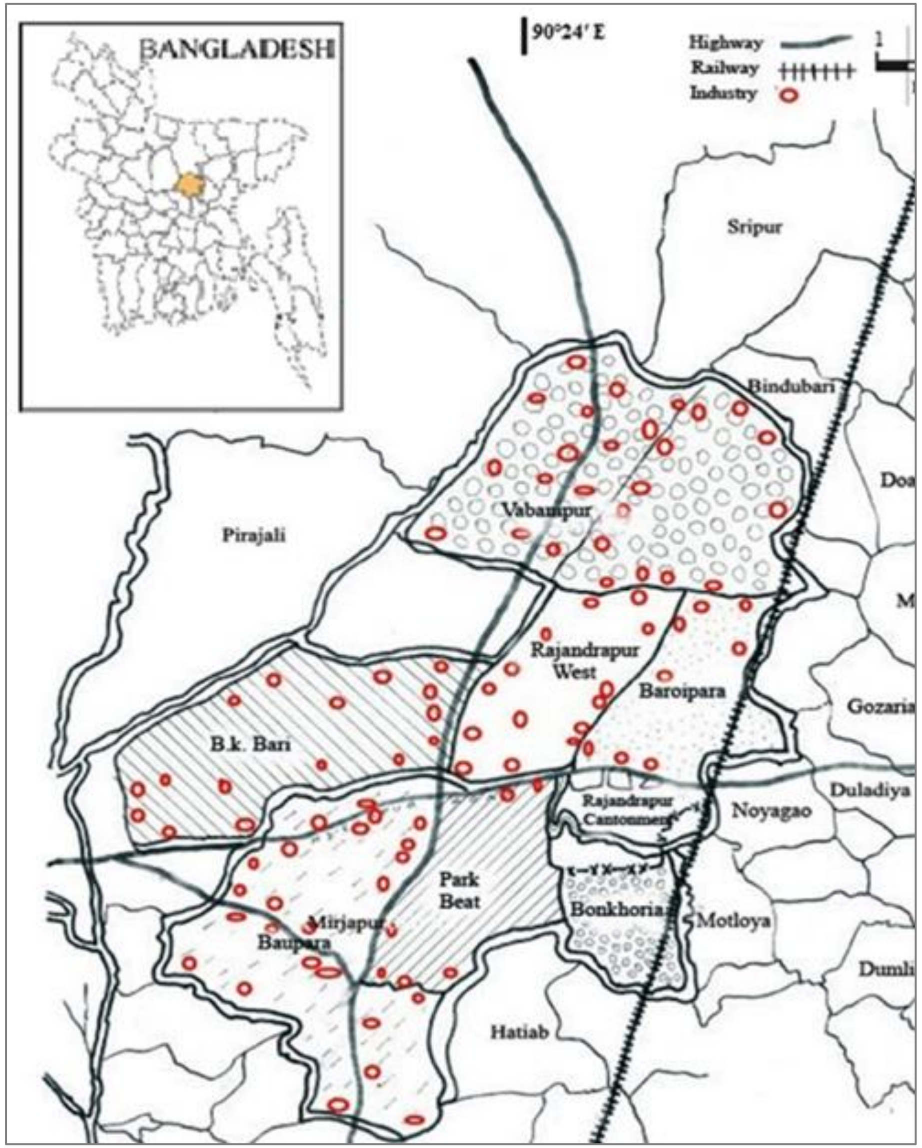
Objective 01: To explore the reasons behind land cover changes of a selected protected area in Bangladesh

❑ Description of the Reasons for Forest Land Cover Change in the Study Area

Growth of Industries within and around Bhawal National Park



Source: BFD, 2024



Source: BFD, 2024

Objective 01: To explore the reasons behind land cover changes of a selected protected area in Bangladesh

Socio-Economic Pressure, Livelihood, and Forest Resource Exploitation

Past Livelihood	Present Livelihood	Remarks
Forest guard for critical habitat	**Drive auto rickshaw** within the forest area	**Critical habitats are destroyed**. No guard is needed now.
Collect forest resource, wood, timber and sell in the market	**Do services in different industries**. No permanent job	**Growth of employment** through industries. Rural non-farm employment increased.
Collect honey in the forest	Involved in **agricultural activities** or agro based small scale farms	**Honey in the forest** area is **extinct** now. **Agricultural** activities **increased**
Collect fish from the river and sell the local market	**Auto rickshaw driver**. Sometimes sell foods in the tourist spot	Fresh water **fish is not found** due to **water pollution.**
Harvest crops **once in a year**	Harvest crops **two or three time** in a year.	**Agricultural** activities **intensified.** Access to micro credits for seeds, fertilizers.
Collect fuel wood from forest, logging, felling of trees	**Homestead vegetation**, poultry, livestock	**Access to micro credits** for farming

- ❑ **Dependency on forest resources has reduced with improved socio economic situation**

Source: Prepared by Author (2024) from FGDs with local people

Source: Field Survey, 2024

Source: Field Survey, 2024

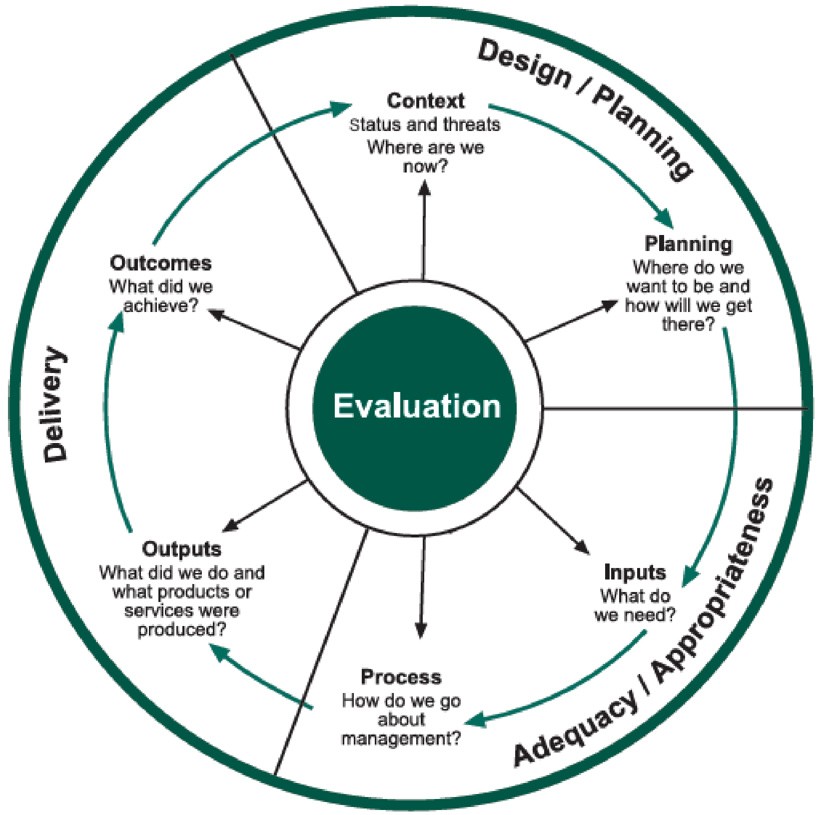
Objective 02

To evaluate management strategies for selected protected area in Bangladesh.

Methodology

❑ IUCN-WCPA framework for assessing management effectiveness of protected areas

	Design		Appropriateness/Adequacy		Delivery	
Elements of management cycle	**Context**	**Planning**	**Inputs**	**Process**	**Outputs**	**Outcomes**
Focus of evaluation	Assessment of importance, threats and policy environment	Assessment of protected area design and planning	Assessment of resources needed to carry out management	Assessment of the way in which management is conducted	Assessment of the implementation of management programmes and actions; delivery of products and services	Assessment of the outcomes and the extent to which they achieved objectives
Criteria that are assessed	Significance/ values Threats Vulnerability Stakeholders National context	Protected area legislation and policy Protected area system design Protected area design Management planning	Resources available to the agency Resources available to the protected area	Suitability of management processes and the extent to which established or accepted processes are being implemented	Results of management actions Services and products	Impacts: effects of management in relation to objectives



Hockings et al., 2006

❑ In **1997**, the **IUCN** World Commission on Protected Areas (WCPA) first **developed** of **a framework** for **evaluating management effectiveness**

❑ In **2000**, **IUCN** developed *the **Management Effectiveness Tracking Tool** (METT)* for applying this framework (Hockings et al., 2006).

❑ This **framework** has been **applied** in **127 countries** by governments, int. organizations or NGOs (UNEP-WCMC, 2024).

Methodology

- Problem Identification
- Formulation of Objective
- Selection of Study Area
- Data Collection
- Data Analysis
- Formulation of Recommendation

Management Effectiveness Tracking Tool (METT)

Understanding context of the study area

Datasheet 01: Input Basic Information of Study Area

Datasheet 02: Threat Assessment

Select the threats present in the study area **from 13 threat categories**

Evaluate their **extent** (whether they occur throughout the entire area or at specific points) and **severity** (impact on study area).

Score the threat **extent and severity levels** from **0 (low) to 3 (very high).**

Calculate % of threat extent and severity to **identify the most damaging** threats.

Management Effectiveness Assessment

Datasheet 03: 38 multiple-choice questions related to **planning, inputs, process, outputs, and outcomes**

Each question has **four scenarios**, scored from 0 (**least desirable) to 3 (most desirable**).

Selects the scenario that best describes the current **situation in the study area** for each question

After completing all questions, a **percentage score is calculated** to determine overall management effectiveness.

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

○ Existing Context of Bhawal National Park

❑ Identification of Possible Threats to Bhawal National Park, Extent and Severity

Detailed assessment of threats

Threat category	Type of Threats	Threat extent	Threat severity
1: Residential and commercial development within a protected area	Housing and settlement	Medium	High
	Commercial and industrial areas	Very high	Very high
	Tourism and recreation infrastructure	Low	Low
2: Threats from farming and grazing as a result of agricultural expansion and intensification	Annual and perennial non-timber crop cultivation	High	High
	Livestock farming and grazing	Low	Medium
	Wood and pulp plantations	Low	Medium
3. Energy production and mining Threats from production of non-biological resources	-	-	-
4. Threats from transport and a range of linear developments,	Tourism and recreation infrastructure	Low	Low
	Roads and railroads	Medium	High
5. Biological resource use and harm	Logging	Low	Medium
	Hunting, killing and collecting terrestrial (native) animals	Medium	High
	Gathering terrestrial (native) plants or plant products (non-timber)	Low	Low
6. Human disturbance	Deliberate vandalism, destructive activities or threats to protected area by staff and visitors	Low	Low
	Recreational activities and tourism	Low	Medium
	Research, education and other activities	Low	Low
	Activities of park authority (e.g. construction or vehicle use, artificial watering points and dams)	Low	Low

Low	Medium	High	Very High

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

- **Existing Context of Bhawal National Park**

❑ Identification of Possible Threats to Bhawal National Park, Extent and Severity

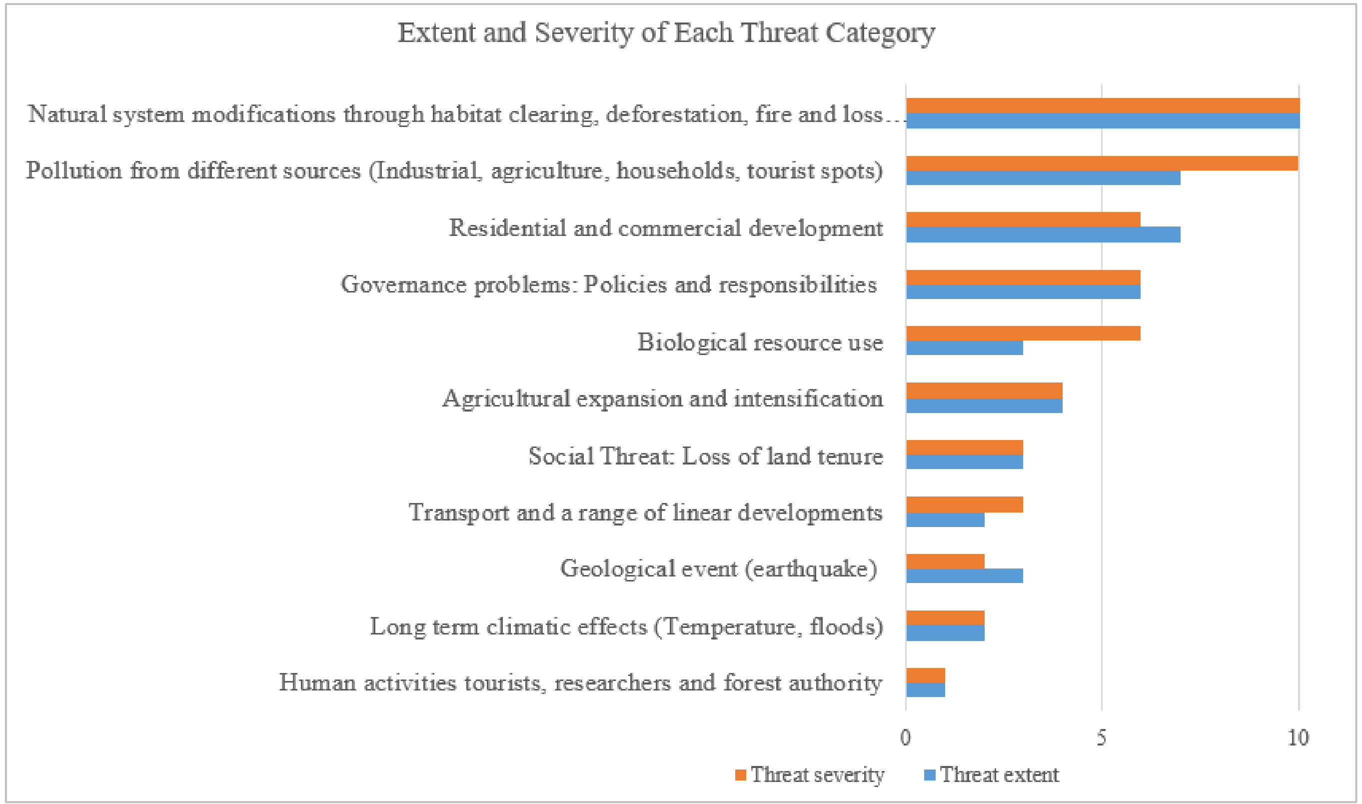
Threat category	Type of Threats	Threat extent	Threat severity
7. Natural system modifications Threats from other actions that convert or degrade habitat or change the way the ecosystem functions.	Loss of keystone species	High	Very high
	Habitat clearing	High	High
	Fire	High	High
	Increased fragmentation	Medium	Very high
9. Pollution entering or generated Threats from introduction of exotic and/or excess materials or energy from point and non-point sources	Garbage and solid waste	Medium	High
	Household sewage and urban waste water	Medium	High
	Sewage and waste water from protected area facilities (e.g. toilets, hotels etc)	Medium	High
	Industrial, mining and military effluents and discharges	Very high	Very high
	Agricultural and forestry effluents (e.g. excess fertilizers or pesticides)	Medium	High
10. Geological events	Earthquakes/Tsunamis	Very high	High
11. Climate change and severe weather	Storms and flooding	Low	Low
	Temperature extremes	High	High
12. Cultural and social threats	Loss of land tenure (including land grabbing)	Very high	Very high
	Loss of cultural links, traditional knowledge and/or management practices and use	Medium	Medium
13. Governance problems	Conflicting policies across sectors	High	Very high
	Confusion about government roles and responsibilities	High	High

Low	Medium	High	Very High

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

○ Existing Context of Bhawal National Park

❑ Understanding the Vulnerability of Bhawal National Park to the Identified Threats



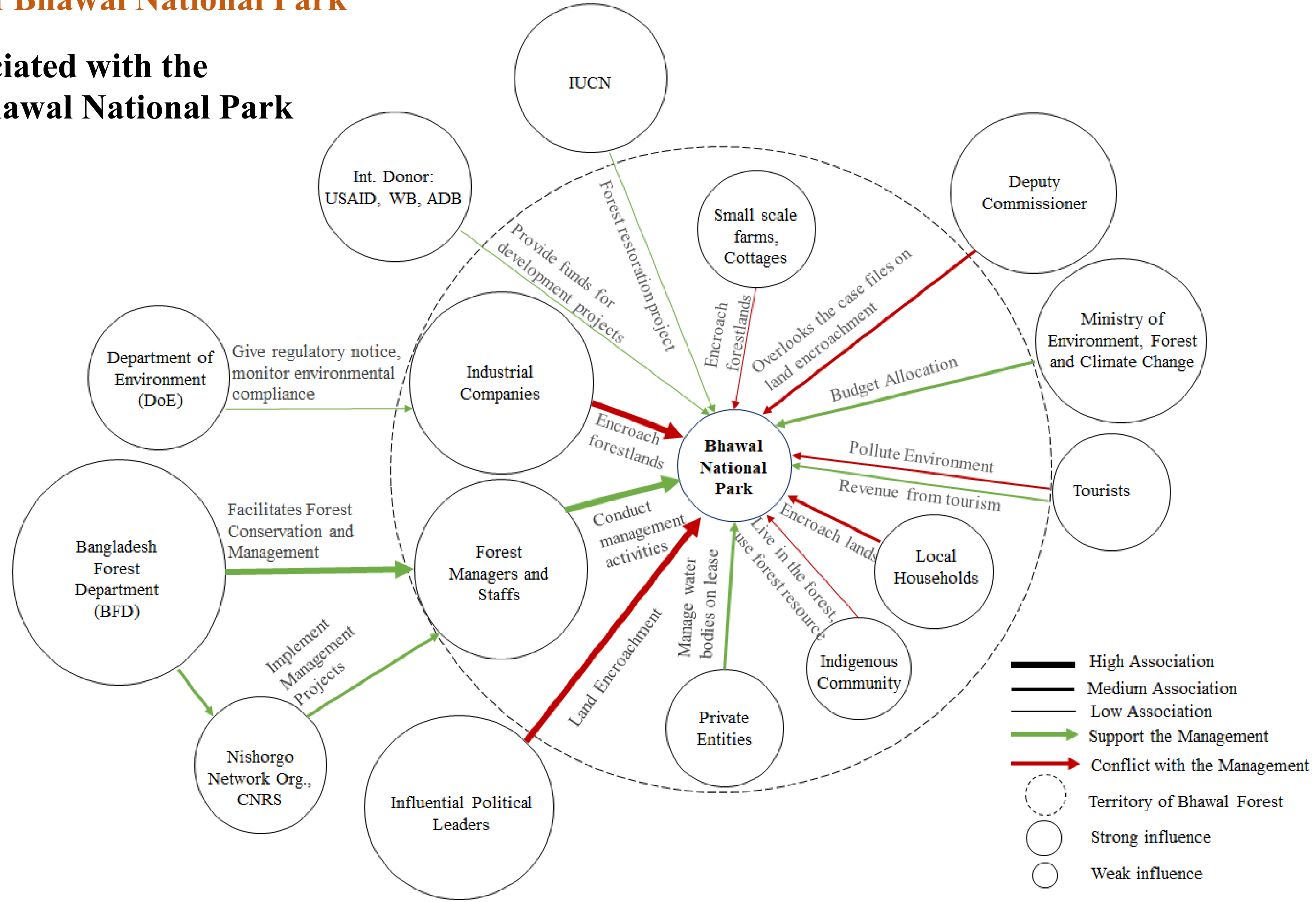
Source: Prepared by Author (2024) from KII with forest managers and FGDs with local

- ❑ **70 percent forest habitats** are **degraded** in the Bhawal National Park (Chowdhury et al., 2013)
- ❑ **Diversity of plant species decrease** in the surrounding areas of human settlements (Jamil et al., 2022)
- ❑ Industries around Bhawal forest areas fall under red category of industry, without any installation of effluent treatment plant (ETP)
- ❑ **Water pollution** in surrounding areas
- ❑ **Destruction of Chilai River** and other water bodies and **Loss of aquatic ecosystem**.
- ❑ Deteriorated water **hampers crop production**, livelihood option (fisheries) of local people

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

- **Existing Context of Bhawal National Park**

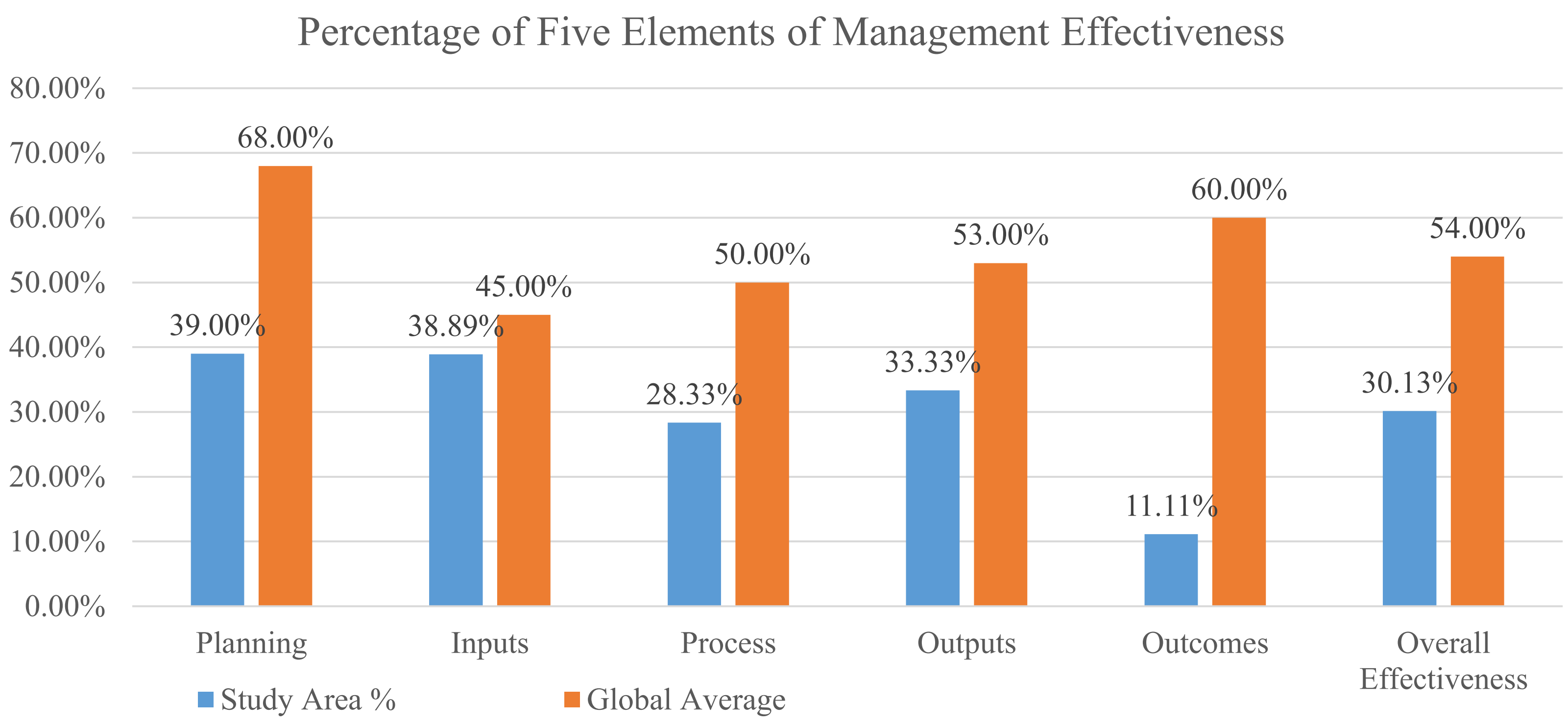
❑ Stakeholders Associated with the Management of Bhawal National Park



Source: Prepared by Author (2024) from KII and FGDs

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

- **Assessment of Management Effectiveness of Bhawal National Park**



Source: Compiled by Author (2024) from METT tool, Leverington et al., 2020

- According to global assessment of 8000 protected areas, the **global average** of management effectiveness is **54 percent** (Leverington et al., 2020).
- Management effectiveness of study area is only **30.13 percent**.
- Significant drawbacks in the management of study area, reducing its conservation efforts to protect the forest ecosystem.

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

○ Management Planning of Bhawal National Park

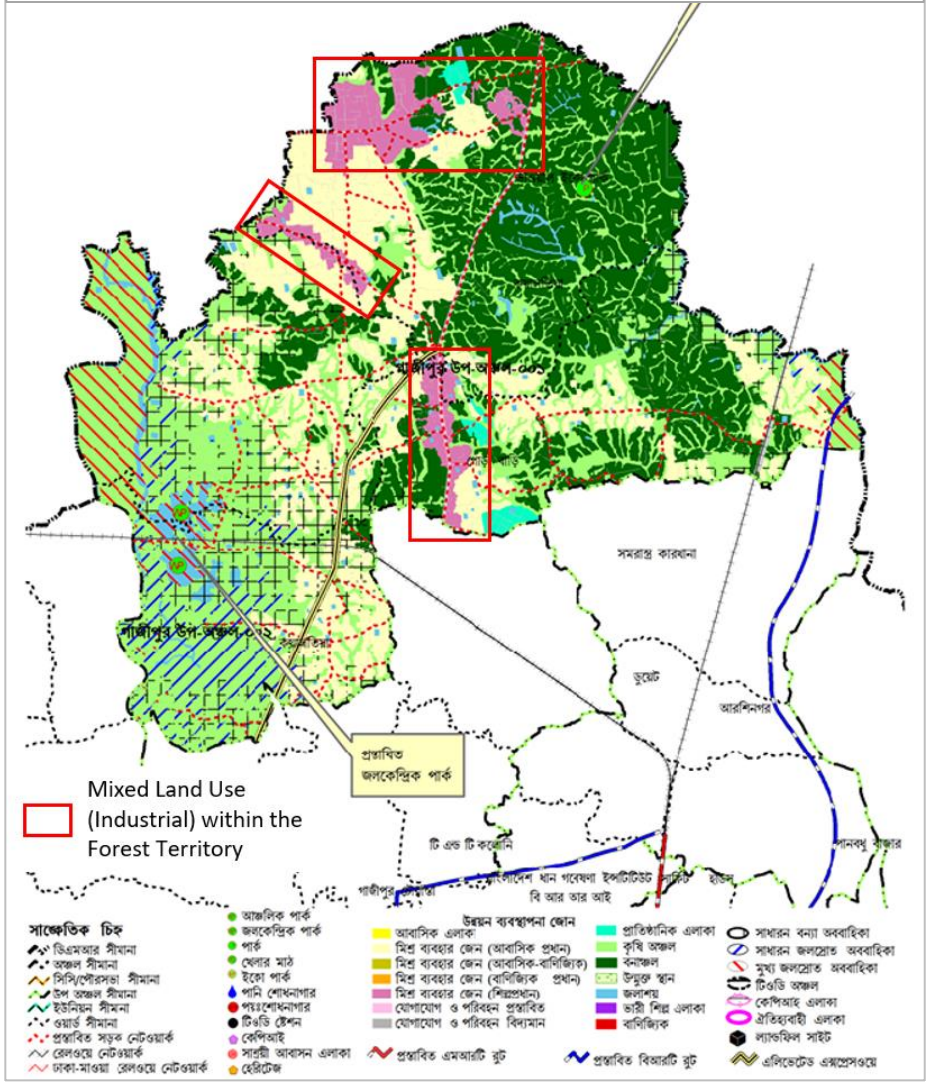
Concerning Issues in Planning	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT score	Max. Score
Legal status	BNP has been **formally gazetted in 1982** as a protected area	Most Desirable	3	3
Appropriate management to achieve the objectives	The protected area has agreed objective but is only partially managed to achieve these objectives.	Least Desirable	0	3
Land use plans recognize/ support this area	Partially supports its conservation issues in land use plans	Moderate	2	3
Rightly designed	Inadequacies in protected area design to achieve its objective	Less Desirable	1	3
Management plan is implemented	**Management plan** has been prepared **in 2014**, but it did **not progress into execution**	Less Desirable	1	3
Allows all stakeholders in management	**Involvement of all stakeholders** is not **ensured**	Least Desirable	0	3
The regular work plan is prepared	**No regular work plan** exists	Least Desirable	0	3
Total Percentage of Management Planning			**47.62%**	100%

Source: Prepared by Author (2024) from KII

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

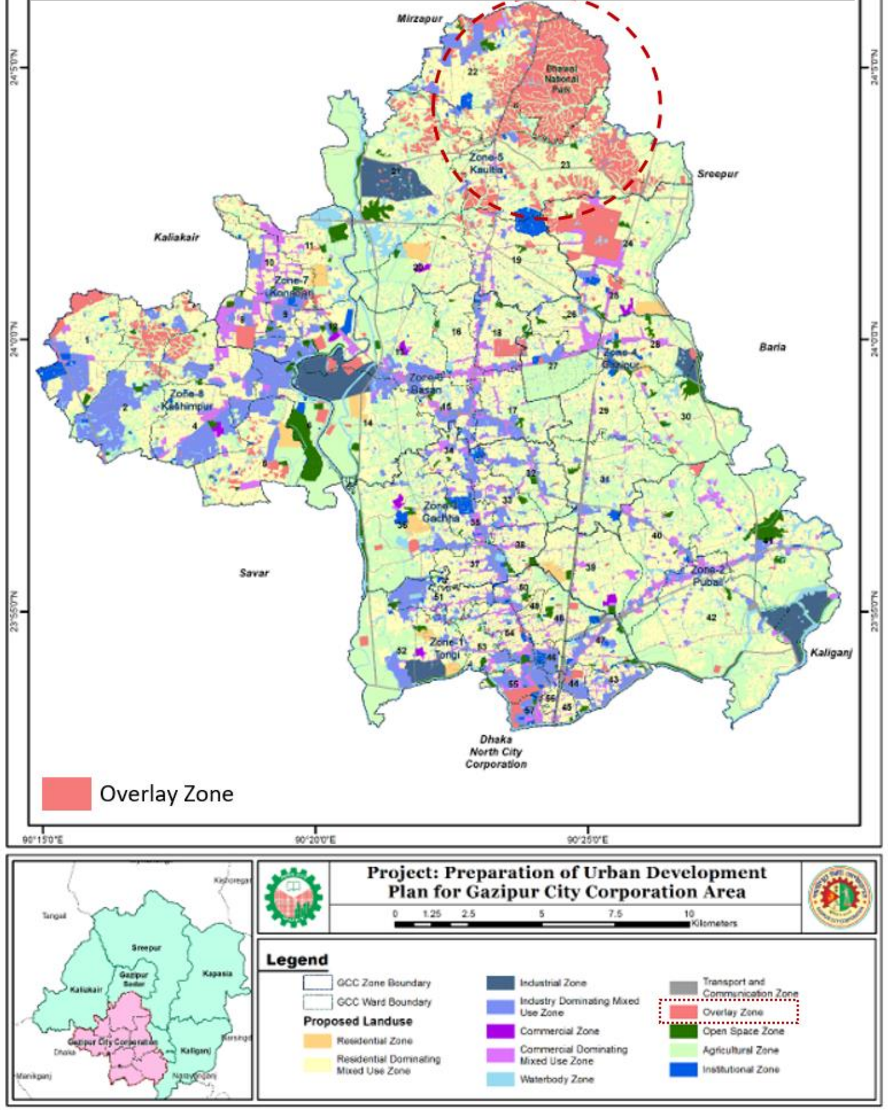
- **No Recognition as a Protected Area in Land Use/ Relevant Plans**

Proposed Land Use of Sub Region 01 of Gazipur District in DAP (2022-2035)



Source: RAJUK, 2023

Proposed Land Use of UDP, 2024 of GCP



Source: CRTS, 2024

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

- **Inputs in the Management of Bhawal National Park**

Concerning Issues in Inputs	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT Score	Max. Score
Information are available to manage	Information on the critical habitats, and cultural values are available to manage some areas	Moderate	2	3
Enough people to manage	**Insufficient number of people** to manage the area	Less desirable	1	3
Proper skills	Knowledge and **skills are low** relative to the needs of the protected area	Less desirable	1	3
Budget	The available **budget is inadequate** for basic management	Less desirable	1	3
The budget is secured	There is very little secured budget and the protected area could not meet its management needs without funding from outside	Less desirable	1	3
Equipment and facilities are sufficient	**Insufficient equipment and facilities** to meet management needs	Less desirable	1	3
Total Percentage of Management Inputs			**38.89%**	100%

Source: Prepared by Author (2024) from KII

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

Process of Management in Bhawal National Park

Concerning Issues in Inputs	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT Score	Max. Score
Regulations and controls	Even though there are **some regulations**, but there exists **major weaknesses**	Less desirable	1	3
Boundary	The **boundary** of the protected area is **not demarcated**	Least desirable	0	3
The budget is managed	**Poor** and constraint **budget management**	Less desirable	1	3
The management staffs can enforce regulation	**Staffs have no enforcement capacity** to enforce protected area legislation and regulations	Least desirable	0	3
Patrolling system	**Patrolling system** can not **control access**	Least desirable	0	3
The staffs have safe working condition	There exists lack of effective measures for providing a safe environment	Less desirable	1	3
Management oriented survey and research work is conducted	Survey and research works are conducted but partially oriented towards the needs of protected area management	Moderate	2	3

Source: Prepared by Author (2024) from KII

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

○ Process of Management in Bhawal National Park

Concerning Issues in Inputs	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT Score	Max. Score
Management activities are monitored, evaluated	There is **no monitoring or evaluation of management activities** in the protected area	Least desirable	0	3
Active resource management	Active **resource management** of critical habitats, species, ecological processes are **not undertaken**	Least desirable	0	3
Climate change consideration	There have **no efforts to consider adaptation** to climate change in management	Least desirable	0	3
Provision of ecosystem services	The ecosystem services delivered by the protected area are understood and some management actions are required	Moderate	2	3
A planned education programme exists linked to the management needs	There is a limited and ad hoc education and awareness programme	Moderate	2	3

Source: Prepared by Author (2024) from KII

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

○ Process of Management in Bhawal National Park

Concerning Issues in Inputs	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT Score	Max. Score
Cooperation with commercial users	There is contact between protected area managers but **no cooperation exits**	Least desirable	0	3
Commercial tour operators contribute to management	Only confined to regulatory matters, not protected area management	Less desirable	1	3
The collected fees are utilized in protected area management	Fees are collected and make substantial contribution to the protected area	Most desirable	3	3
Indigenous people are involved in management	**Indigenous people are not involved in management**	Least desirable	0	3
Local communities living in or near the protected area have input in management decision	**Local communities are not involved in management** decision	Least desirable	0	3
Total Percentage of Management Process			**38.89%**	100%

Source: Prepared by Author (2024) from KII

Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

- ### Outputs from the Management of Bhawal National Park

Anticipated outputs from protected area	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT Score	Max. Score
Visitor facilities and services are adequate and maintained	There are visitor facilities but proper maintenance is not ensured	Moderate	2	3
The protected area provide sustainable livelihood benefits to local communities and indigenous people	BNP provides livelihoods to the people but not in a sustainable way	Moderate	2	3
Threats to the natural values are effectively assessed	Threats are assessed but not in detail or effective means	Less desirable	1	3
Functional connectivity is ensured	**Functional connectivity** of the protected area is **not ensured**	Least desirable	0	3
Total Percentage of Outputs From BNP			33.33%	100%

Source: Prepared by Author (2024) from KII

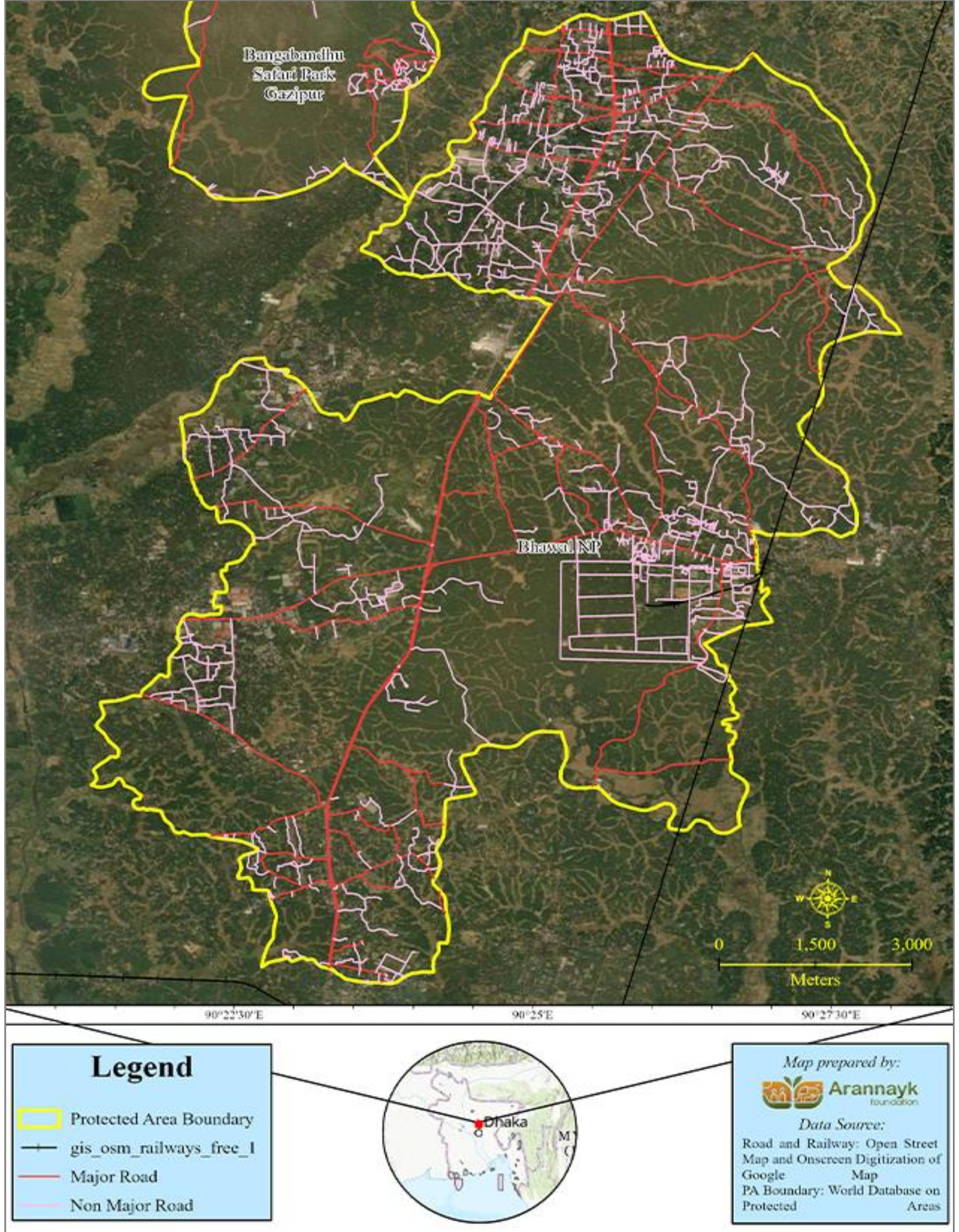
Objective 02: To evaluate management strategies for selected protected area in Bangladesh.

- **Outcomes from the Management of Bhawal National Park**

Anticipated outcomes from protected area	Current Scenario in Bhawal National Park (BNP)	Level of Condition	METT Score	Max. Score
Condition of important natural values	**Some natural values are severely damaged**	Less desirable	1	3
Condition of key species is stable over the years	The key species of BNP have degraded over the years	Least desirable	0	3
Conservation status of habitats is in place	**Conservation status of habitats has worsened** over the years	Least desirable	0	3
Total Percentage of Outputs From BNP			**11.11 %**	100%

Source: Prepared by Author (2024) from KII

- Forest Fragmentation in the Study Area due to Road Networks



Source: Arannayk Foundation, 2022

Major Findings

- The forestlands of Bhawal National Park are gradually encroached through establishing industries, settlements or expanding agricultural lands.
- At present, **989.36 acres forestlands are illegally encroached** by industries, agro based farms or private cottages. The total land valuation of the **encroached lands is 197872 crore BDT.**
- The Forest Department have filed **985 eviction cases** against the illegal encroachers. Only **326.36 acres (30%) forestlands are restored from the encroachers.**
- Before 2000, the issue of forest resource exploitation was extremely high as the forest resources were the only means of living for the poor people.
- The management effectiveness of the protected area in only 30.13 percent. The present management system of the protected area is ineffective is conserving forest ecosystem.
- **Bangladesh Forest Department has weak institutional capacity** to directly enforce the laws against the forest crimes.
- The current management process is **excluding the local people from decision-making process** and management activities.

Recommendation

1. Addressing Inconsistent Policies and Plans: Forestry Master Plan aims to protect Sal forests, but existing land use plans, DAP (2022-2035) and UDP, 2024 of Gazipur, don't align.
Wildlife Act 2012 restricts new **industries within 2 km**, but **exceptions for existing ones**.

2. Law Enforcement Focus:

- Forest officials need direct enforcement power to combat illegal land encroachment.
- Divisional Forest Office (DFO) should execute Wildlife Act for conservation.
- Political commitment and transparency are crucial for effective enforcement.

3. Central Committee for Monitoring:

- Establish a central committee led by Wildlife Management Division.
- Regularly evaluate management effectiveness using IUCN framework (every 3-5 years).
- Develop conservation strategies, involve locals, and manage financing.

4. Management Plan Implementation

- Involve local communities in implementing the plan
- Seek international donor support to address budget constraints.

5. Collaboration and Capacity Building:

- Engage NGOs, agencies, and communities.
- Focus on capacity development for forest managers through targeted training.

Recommendation

6. Co-Management of the Study Area and Alternate Livelihood Opportunities for the Locals and Tribal Communities

Co-Management Framework:

1. Develop a co-management framework for the study area.
2. Form a **Village Conservation Forum (VCF)** at the village level.
3. VCF educates locals about their roles in sustainable forest management.
4. **Multiple VCFs can be established in different communities** around the protected area.
5. **Create a Co-Management Committee (CMC) with leaders from each VCF**.
6. **Include stakeholders like local people, forest managers, NGOs, and private sectors**.
7. CMCs function independently and share revenue generated within the protected area.

Alternative Livelihood Options:

1. **Identify livelihood alternatives** to reduce overexploitation.
2. **Explore small trades, capacity building, and employment opportunities**.
3. **Involve local people and tribal communities through participatory practices**.
4. Learn from successful examples, such as forestry projects in India.
5. Sustainable livelihood options benefit both people and forest conservation.

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Thank You